

Do Large Language Models (LLMs) Reason Like Us? Even Better?

Hanna Dettki^{1,3}, Brenden Lake^{1,2}, Charley M. Wu^{3,4,5,6}, Bob Rehder¹

1) Department of Psychology, NYU; 2) Center for Data Science, NYU; 3) Department of Computer Science, University of Tübingen; 4)Tübingen AI Center 5) TU Darmstadt; 6) Hessian.ai

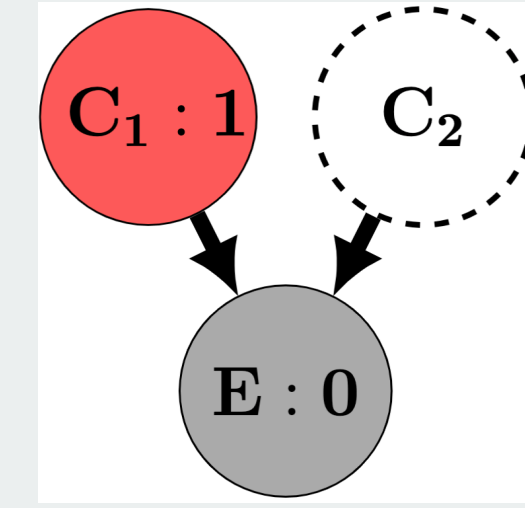


NYU

LLMs Can Do Causal Reasoning*

- We compared LLMs and humans on causal inference judgments.

*on our dataset



“What is probability that the red node is 1, given that we know E is absent and we know nothing about C2 ?”

LLMs Can Predict Human Judgements

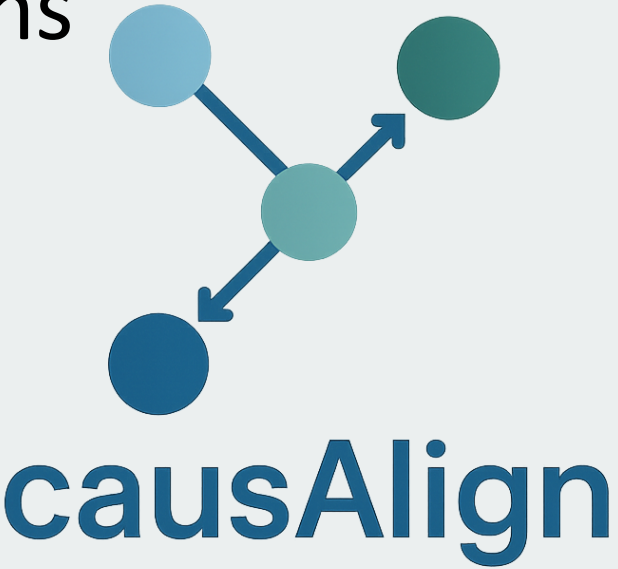
- All LLMs tested (except GPT-3.5) have decent alignment with humans (r=.66-.76)
- Gemini-1.5-pro is most aligned with humans overall (r=.76).

When LLMs Reason Better than Humans

- LLMs don’t fall for the “rich-get-richer” effect that makes humans incorrectly link independent causes (except GPT-3.5)
- Some LLMs correlate better with *ideal causal reasoning*** : (r=.88) vs. (r=.77) for humans

LLMs still Miss subtle Causal Patterns

- e.g., explaining away
- But so do humans



causAlign

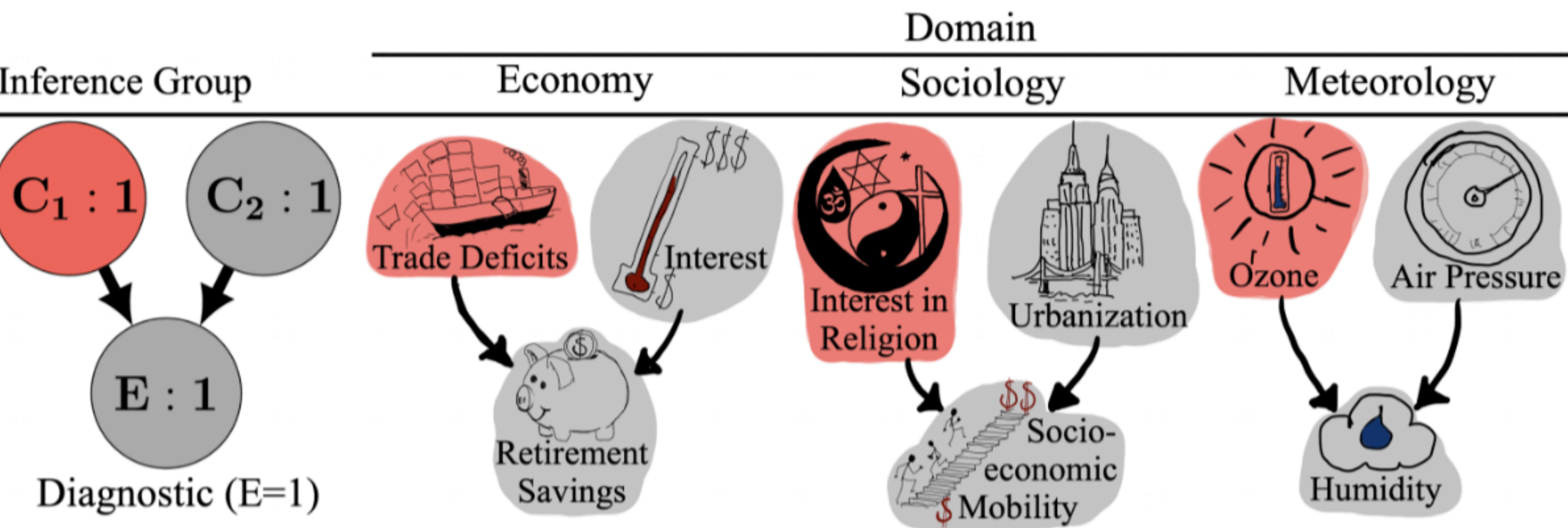
Relevance

- As AI-systems increasingly assist human decision-making, understanding their causal reasoning biases is critical for their safe deployment and reliability.

Approach

- Using Cognitive Science methods to compare the ability of humans and LLMs to perform causal inference on the same tasks.

Inference Tasks



Variable/Node States:

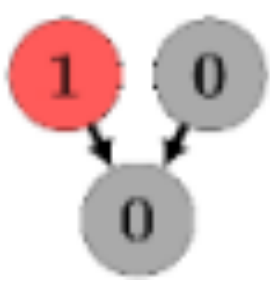
●: latent ●: Observed $\in \{0,1\}$ ○: No information

Example Prompt adapted from [1]:

Domain introduction: Sociologists seek to describe and predict the regular patterns of societal interactions. To do this, they study some important variables or attributes of societies. They also study how these attributes are responsible for producing or causing one another.

- **Causal mechanism:** Assume you live in a world that works like this:
 - * **C1 $\rightarrow$ E:** *High* urbanization causes *high* socio-economic mobility.
 - **Explanation:** Big cities provide many opportunities for financial and social improvement.
 - * **C2 $\rightarrow$ E:** Also, *low* interest in religion causes *high* socio-economic mobility.
 - **Explanation:** Without the restraint of religion-based morality, the impulse toward greed dominates and people tend to accumulate material wealth.
- **Observation:** Now suppose you observe the following: *low* socio-economic mobility and *low* urbanization.

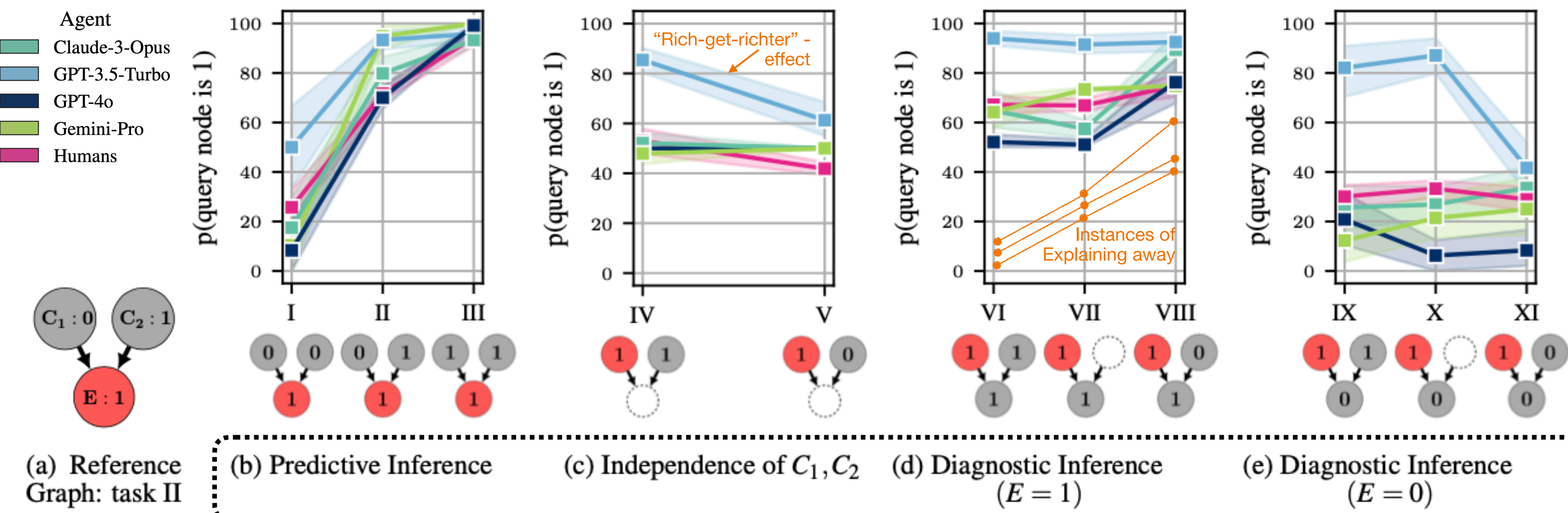
• Inference task, here XI:



Given the observations and the causal mechanism, how likely on a scale from 0 to 100 is *high* urbanization? 0 means definitely not likely and 100 means definitely likely. Please provide only a numeric response and no additional information.

Overview of all Inference Tasks (I-XI) and their Graph Templates

(across all domains; 95% CI via Bootstrapping)



References:

[1] Rehder, B., & Waldmann, M. R. (2017). Failures of explaining away and screening off in described versus experienced causal learning scenarios. *Memory & Cognition*, 45, 245–260.

[2] Zečević, M., Willig, M., Dhama, D.S. and Kersting, K., 2023. Causal parrots: Large language models may talk causality but are not causal. *arXiv preprint arXiv:2308.13067*.



Paper

← Inference Groups